Skybox Mobile API

Reference for the mobile app: authentication, reception, storage and preparation orders (pickers)

Item	Value
Server	Odoo 18 — module **skybox**
Protocol	HTTPS, JSON-RPC 2.0 (`POST`, `Content-Type: application/json`)
Base URL	`https://<odoo-host>`
Authentication	Login API key + user credentials, then a Bearer access token
Session	Token valid 5 minutes, renewed on every call (sliding expiry)
Version	October 2026

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1. Overview and conventions

All mobile endpoints are Odoo **JSON-RPC** routes. Every call is an HTTP `POST` with a JSON body. The business parameters go in the `params` object; the answer is in `result`.

Request envelope

```
POST /skybox/api/<endpoint>
Content-Type: application/json
Authorization: Bearer <access_token>      (all endpoints except login)

{
  "jsonrpc": "2.0",
  "method": "call",
  "id": 1,
  "params": { ...endpoint parameters... }
}
```

Response envelope

```
{
  "jsonrpc": "2.0",
  "id": 1,
  "result": { "success": true, ... }
}
```

Conventions

- The HTTP status is **200** for business errors. Always test `result.success`. On failure the result is `{"success": false, "error": "<code>"}`, sometimes with extra fields (`message`, `state`, ...).
- An `"error"` key at the top level of the envelope (instead of `result`) means a server crash or malformed request (HTTP 200 JSON-RPC error, or HTTP 4xx/5xx).
- Dates are UTC strings `"YYYY-MM-DD HH:MM:SS"`.
- Empty values are returned as `null`.
- The token can also be sent in the body as `params.access_token` instead of the `Authorization` header.

2. Authentication and session

- **Login API key**: each app installation sends an API key of type *Login* (created in Odoo: Skybox › Configuration › API management). It identifies the calling program.
- **Access token**: `/skybox/api/login` returns an `access_token`. Send it as `Authorization: Bearer <token>` on every other call.
- **Sliding expiry — 5 minutes**: each successful call pushes the expiry 5 minutes further. After 5 minutes without any call the token is revoked; the app must log in again. The logout is written in the user's login history ("Session expirée").
- **Heartbeat**: while the app is open, call `/skybox/api/orders` (pickers) or `/skybox/api/me` every **60 seconds**.
- **Roles**: `admin`, `manager`, `agent`, `picker`. A user without a Skybox role gets `access_denied` on business endpoints.
- **Brute-force protection**: after 5 failed logins (same IP + login) the login is blocked for 15 minutes (`too_many_attempts` with `retry_after` seconds).
- Every login and logout (app or Odoo) is recorded in the user's *Historique de connexion* with date, time, IP and connection type.

3. Endpoints — Session

POST `/skybox/api/login` — *Log in and get an access token*

Checks the Login API key and the Odoo user's credentials, then issues an access token. For a **picker**, the login also puts the user in the order assignment cycle and gives them their share of waiting orders.

Authentication: Login API key (body `api_key` or header `X-API-KEY`). No Bearer token.

Parameters (inside `params`)

Name	Type	Req.	Description
api_key	string	yes*	Login API key. *Or header `X-API-KEY`.
email	string	yes	User login / e-mail (alias: `login`).
password	string	yes	User password.

Example request

```
{"jsonrpc": "2.0", "method": "call", "id": 1,
 "params": {"api_key": "sk_login_xxx",
            "email": "picker1@example.com",
            "password": "secret"}}
```

Example result

```
{"success": true,
 "access_token": "q3V9...48 chars...",
 "expiry": "2026-10-12 09:05:00",
 "uid": 6, "name": "Picker 1",
 "login": "picker1@example.com", "email": "picker1@example.com",
 "role": "picker"}
```

Errors: `invalid_api_key`, `missing_credentials`, `too_many_attempts`, `invalid_credentials`

POST `/skybox/api/me` — *Current user / heartbeat*

Returns the user of the token and renews the token for 5 minutes. Use it at app start to check a saved token, and as a heartbeat for non-picker users.

Authentication: Bearer token.

Example request

```
{"jsonrpc": "2.0", "method": "call", "id": 2, "params": {}}
```

Example result

```
{"success": true, "uid": 6, "name": "Picker 1",
 "login": "picker1@example.com", "email": "picker1@example.com",
 "role": "picker", "expiry": "2026-10-12 09:10:00"}
```

Errors: `invalid_token`

POST `/skybox/api/logout` — *Log out*

Revokes the token. For a picker: leaves the assignment cycle, and their unfinished preparation orders are redistributed to the other connected pickers.

Authentication: Bearer token.

Example request

```
{"jsonrpc": "2.0", "method": "call", "id": 3, "params": {}}
```

Example result

```
{"success": true}
```

- Always returns `success: true`, even for an unknown token.

4. Endpoints — Reception and storage

POST `/skybox/api/receive` — *Receive a parcel into a bag*

Creates an Odoo reception for a scanned tracking number, placed in a bag. The agent is the token user; the step is written in the parcel history.

Authentication: Bearer token (any Skybox role).

Parameters (inside `params`)

Name	Type	Req.	Description
tracking_number	string	yes	Carrier tracking number of the parcel.
bag	string	yes*	Bag name (e.g. `BAG001`). *Or `bag_id`.
bag_id	integer	no	Bag id, instead of `bag`.
courier	string	no	Carrier name (e.g. DHL); created if unknown.
notes	string	no	Free notes.

Example request

```
{"jsonrpc": "2.0", "method": "call", "id": 4,
 "params": {"tracking_number": "1Z999AA10123456784",
            "bag": "BAG001", "courier": "UPS"}}
```

Example result

```
{"success": true, "picking_id": 57, "reference": "WH/IN/00057",
 "tracking_number": "1Z999AA10123456784",
 "bag": "BAG001", "bag_id": 3,
 "agent": "Agent 1", "agent_id": 5,
 "warehouse": "Main Warehouse",
 "date": "2026-10-12 09:01:12"}
```

Errors: `invalid_token`, `access_denied`, `missing_tracking_number`, `missing_bag`, `bag_not_found`, `already_received (+ picking_id, reference)`, `no_receipt_type`

POST `/skybox/api/bag` — *Scan a bag: list its parcels*

Returns the parcels inside a bag: pre-alerts assigned to the bag by Skybox plus receptions still in the bag.

Authentication: Bearer token (any Skybox role).

Parameters (inside `params`)

Name	Type	Req.	Description
bag	string	yes*	Bag name. *Or `bag_id`.
bag_id	integer	no	Bag id.

Example request

```
{"jsonrpc": "2.0", "method": "call", "id": 5,
 "params": {"bag": "BAG001"}}
```

Example result

```
{"success": true, "bag": "BAG001", "bag_id": 3,
 "skybox_status": "OPEN", "count": 2,
 "colis": [
   {"tracking_number": "1Z999AA10123456784",
    "prealerte": "PRE/0012", "prealerte_id": 12,
    "processing_state": "not_processed",
    "skybill": null, "suite_number": "SKY-1042",
    "client": "John Doe", "courier": "UPS", ...},
   ...]}
```

Errors: `invalid_token`, `access_denied`, `missing_bag`, `bag_not_found`

POST `/skybox/api/store` — *Store a parcel in a location*

Moves a parcel of a bag (status Receipt or Processed) into a storage location; its status becomes *Stored*. This is the only route that stores a parcel.

Authentication: Bearer token (any Skybox role).

Parameters (inside `params`)

Name	Type	Req.	Description
tracking_number	string	yes	Tracking number of the parcel.
bag	string	yes*	Bag name. *Or `bag_id`.
location	string	yes*	Location barcode, full name (`WH/Stock/Zone A1`) or name. *Or `location_id`.
location_id	integer	no	Internal location id.

Example request

```
{"jsonrpc": "2.0", "method": "call", "id": 6,
 "params": {"tracking_number": "1Z999AA10123456784",
            "bag": "BAG001", "location": "WH/Stock/Zone A1"}}
```

Example result

```
{"success": true, "bag": "BAG001", "bag_id": 3,
 "location": "WH/Stock/Zone A1", "location_id": 21,
 "agent": "Agent 1", "agent_id": 5,
 "date": "2026-10-12 09:04:40", "count": 1,
 "orders": [{"picking_id": 57, "reference": "WH/IN/00057",
             "tracking_number": "1Z999AA10123456784",
             "client": "John Doe"}]}
```

Errors: `invalid_token`, `access_denied`, `missing_tracking_number`, `missing_bag`, `bag_not_found`, `missing_location`, `location_not_found`, `order_not_in_bag`

5. Endpoints — Preparation orders (pickers)

Preparation orders are sent by Skybox to Odoo and assigned automatically to connected pickers (see section 6). A picker only sees and changes **their own** orders; managers and admins can open any order. Steps done in the app are sent to Skybox exactly like the buttons in Odoo.

Order statuses

state	Label	Done from
`new`	Nouveau	Created by Skybox
`picking`	Picking	App (step or first scan) / Odoo
`picked`	Picked	App / Odoo
`packed`	Packed	App / Odoo
`ready`	Prêt à expédier	App / Odoo
`dispatched`	Expédié	Odoo only (carrier + tracking)
`cancelled`	Annulé	Skybox

Steps only move forward; a step can skip intermediate ones (e.g. `new` → `packed`).

POST **`/skybox/api/orders`** — *My open orders (also the heartbeat)*

Lists the open orders (new → ready) assigned to the token user, oldest first. Call it every 60 seconds while the app is open: it refreshes the list and keeps the picker in the assignment cycle.

Authentication: Bearer token.

Parameters (inside `params`)

Name	Type	Req.	Description
state	string	no	Filter on one status, e.g. `new`.

Example request

```
{"jsonrpc": "2.0", "method": "call", "id": 7, "params": {}}
```

Example result

```
{"success": true, "count": 1,
 "orders": [
   {"order_id": 12, "order": "PO-2026-0012", "reference": "PREP/00012",
    "state": "new", "state_label": "Nouveau",
    "priority": "HIGH", "type": "STANDARD",
    "client": "John Doe", "suite_number": "SKY-1042",
    "skybill_number": "SB-88231",
    "total_parcels": 3, "picked_parcels": 0, "total_weight": 4.25,
    "requested_at": "2026-10-12 08:55:00",
    "has_exception": false, "sync_state": "none"}]}
```

Errors: `invalid_token`, `access_denied`

POST **`/skybox/api/order`** — *Order detail with parcels to pick*

Full order: delivery address, instructions, allowed next steps and every parcel with its location in the warehouse.

Authentication: Bearer token.

Parameters (inside `params`)

Name	Type	Req.	Description
order_id	integer	yes*	Order id. *Or `order`.
order	string	no	Order number or reference, instead of `order_id`.

Example request

```
{"jsonrpc": "2.0", "method": "call", "id": 8,
 "params": {"order_id": 12}}
```

Example result

```
{"success": true,
 "order_id": 12, "order": "PO-2026-0012", "state": "new", ...
 "instructions": "Fragile, double box",
 "warehouse": "Main Warehouse", "package_code": null,
 "ship_to": {"name": "John Doe", "phone": "+212600000000",
             "address": "12 Rue X", "city": "Casablanca",
             "country": "Morocco"},
 "next_steps": ["picking", "picked", "packed", "ready"],
 "lines": [
   {"line_id": 31, "tracking_number": "1Z999AA10123456784",
    "skybill_number": "SB-88231", "bag": "BAG005_OUT",
    "zone": "A", "aisle": "03", "rack": "R2",
    "location": "WH/Stock/Zone A1",
    "weight": 1.2, "fragile": false, "pick_state": null},
   ...]}
```

Errors: `invalid_token`, `access_denied`, `missing_order`, `order_not_found`, `not_your_order`

- The summary fields of `/orders` are also included.

POST `/skybox/api/order/scan` — *Scan a parcel during picking*

Marks one parcel of the order as picked, missing or damaged. If the order is still *Nouveau*, the first scan moves it to *Picking* (sent to Skybox).

Authentication: Bearer token.

Parameters (inside `params`)

Name	Type	Req.	Description
order_id	integer	yes*	Order id. *Or `order`.
tracking_number	string	yes	Scanned tracking number (case-insensitive).
pick_state	string	no	`PICKED` (default), `MISSING` or `DAMAGED`.

Example request

```
{"jsonrpc": "2.0", "method": "call", "id": 9,
 "params": {"order_id": 12,
            "tracking_number": "1Z999AA10123456784",
            "pick_state": "PICKED"}}
```

Example result

```
{"success": true, "order_id": 12, "state": "picking",
 "total_parcels": 3, "picked_parcels": 1, ...
 "line_id": 31, "tracking_number": "1Z999AA10123456784",
 "pick_state": "PICKED"}
```

Errors: `invalid_token`, `access_denied`, `missing_order`, `order_not_found`, `not_your_order`, `missing_tracking_number`, `invalid_pick_state`, `order_not_in_picking (+ state)`,

`parcel_not_in_order`, `step_refused (+ message)`

- Scanning is allowed only while the order is `new` or `picking`.
- Scanning the same parcel again overwrites its `pick_state`.

POST **`/skybox/api/order/step`** *Move the order to a next step*

Moves the order forward and sends the new status to Skybox. Same rules as the Odoo buttons: `picked` marks all unscanned parcels as PICKED; `packed` sets the box code (order reference by default).

Authentication: Bearer token.

Parameters (inside `params`)

Name	Type	Req.	Description
order_id	integer	yes*	Order id. *Or `order`.
step	string	yes	`picking` | `picked` | `packed` | `ready`.
package_code	string	no	Final box code, used with `packed`.

Example request

```
{"jsonrpc": "2.0", "method": "call", "id": 10,
 "params": {"order_id": 12, "step": "packed",
            "package_code": "CART-0012"}}
```

Example result

```
{"success": true, "order_id": 12, "state": "packed",
 "state_label": "Packed", ...
 "sync_state": "sent",
 "sync_message": "<Skybox answer>"}
```

Errors: `invalid_token`, `access_denied`, `missing_order`, `order_not_found`, `not_your_order`, `invalid_step (+ allowed)`, `step_not_allowed (+ state)`, `step_refused (+ message)`

- `sync_state`: `sent` = accepted by Skybox; `pending` = Skybox unreachable, Odoo retries automatically every 5 minutes; `error` = refused by Skybox (see `sync_message`). The step is saved in Odoo in all cases.
- Dispatch (carrier + carrier tracking number) and exceptions are done in Odoo.

6. Picker assignment cycle

Each new preparation order is assigned automatically, in turn (round-robin), to the pickers currently in the cycle.

- **Who is in the cycle**: users with the *Picker* role (not agents, managers or admins) who have a valid token used during the last **5 minutes**. Order of the cycle = order of login.
- **Example**: 3 pickers logged in (A, B, C). Orders 1–5 go to A, B, C, A, B. Order 6 goes to C. If picker D logs in, D joins the end of the cycle and gets order 7.
- **Logout** (or 5 minutes without any call): the picker leaves the cycle and their open orders (new → ready) are redistributed one by one to the remaining pickers.
- **Nobody connected**: orders wait (no picker) and are distributed as soon as a picker logs in.
- **Several devices**: a picker logged in on two devices has a single place in the cycle and keeps their orders while one device is still active.
- Each login, logout and redistribution is visible in Odoo on the user form (*Historique picker* tab), with a warning when a picker left with active orders.

7. Recommended app flows

Picker

```
1. POST /skybox/api/login            -> save access_token, check role == "picker"
2. every 60 s: POST /skybox/api/orders   (refresh list + heartbeat)
3. tap an order: POST /skybox/api/order {order_id}
4. for each parcel scanned:
       POST /skybox/api/order/scan {order_id, tracking_number}
5. POST /skybox/api/order/step {order_id, step: "picked"}
6. POST /skybox/api/order/step {order_id, step: "packed", package_code}
7. POST /skybox/api/order/step {order_id, step: "ready"}
8. on exit: POST /skybox/api/logout
```

Reception agent

```
1. POST /skybox/api/login
2. scan parcel + bag:   POST /skybox/api/receive {tracking_number, bag}
3. scan bag:            POST /skybox/api/bag {bag}
4. store each parcel:   POST /skybox/api/store {tracking_number, bag, location}
5. idle screens: POST /skybox/api/me every 60 s
```

Handling invalid_token

On `invalid_token` from any endpoint, clear the saved token and show the login screen. Do not retry the call with the same token.

8. Error code reference

Code	Endpoints	Meaning / what to do
invalid_api_key	login	Login API key missing, wrong or inactive.
missing_credentials	login	email/login or password missing.
too_many_attempts	login	Blocked after 5 failures; wait `retry_after` seconds.
invalid_credentials	login	Wrong login or password.
invalid_token	all except login	Token missing, expired (5 min idle) or logged out. Log in again.
access_denied	business endpoints	User has no Skybox role.
missing_tracking_number	receive, store, order/scan	tracking_number missing.
missing_bag / bag_not_found	receive, bag, store	Bag not sent / unknown.
already_received	receive	Tracking already received; returns picking_id and reference.
no_receipt_type	receive	No reception operation type configured for the user's warehouse.
missing_location / location_not_found	store	Location not sent / unknown internal location.
order_not_in_bag	store	No parcel with this tracking in the bag (status Receipt/Processed).
missing_order / order_not_found	order, order/scan, order/step	order_id/order not sent / unknown.
not_your_order	order, order/scan, order/step	Order assigned to another picker.
invalid_pick_state	order/scan	pick_state not PICKED, MISSING or DAMAGED.
order_not_in_picking	order/scan	Order is past the picking stage (returns state).
parcel_not_in_order	order/scan	Scanned parcel does not belong to this order.
invalid_step	order/step	Unknown step (returns allowed).
step_not_allowed	order/step	Step not after the current state, or order finished (returns state).
step_refused	order/scan, order/step	Business rule refused the step; show `message`.